

Scattering of Light

Kai Cui

1 Molecular Polarizability

The polarizability of a molecule describes how the molecular dipole response to the external electric field. Let $\hat{H}_0$ be the Hamiltonian of the free molecule. E_a and $|a\rangle$ are the eigenvalue and eigenstates of $\hat{H}_0$,

$$\hat{H}_0|a\rangle = E_a|a\rangle \quad (1)$$

The time evolution operator for the free molecule is given by

$$\hat{U}_0(t) = e^{-it\frac{\hat{H}_0}{\hbar}} \quad (2)$$

Using the dipole approximation, an external electric field interact with the molecule via

$$\hat{H}_1(t) = -\hat{\boldsymbol{\mu}} \cdot \mathbf{E}(t) \quad (3)$$

where $\hat{\boldsymbol{\mu}}$ is the dipole operator. In this section, the electric field is treated classically. We also assume that the wavelength of the light is much larger than the size of the molecule thus $\mathbf{E}(t)$ is uniform in space. The time evolution operator of this perturbed system is

$$\hat{U}(t) = e^{-it\frac{\hat{H}_0}{\hbar}} \hat{T} \exp \left\{ \frac{i}{\hbar} \int_0^t dt' \hat{\boldsymbol{\mu}}_0(t') \cdot \mathbf{E}(t') \right\} \quad (4)$$

where $\hat{T}$ is time ordering operator,

$$\hat{\boldsymbol{\mu}}_0(t) \equiv \hat{U}_0^\dagger(t) \hat{\boldsymbol{\mu}} \hat{U}_0(t) = e^{it\frac{\hat{H}_0}{\hbar}} \hat{\boldsymbol{\mu}} e^{-it\frac{\hat{H}_0}{\hbar}} \quad (5)$$

In linear response theory, we can approximate $\hat{U}(t)$ to its first-order, which is

$$\hat{U}(t) \approx e^{-it\frac{\hat{H}_0}{\hbar}} \left(1 + \frac{i}{\hbar} \int_0^t dt' \hat{\boldsymbol{\mu}}_0(t') \cdot \mathbf{E}(t') \right) \quad (6)$$

In Heisenberg's picture, the time-dependent dipole operator $\hat{\boldsymbol{\mu}}(t)$ in the perturbed system is given by

$$\hat{\boldsymbol{\mu}}(t) = \hat{U}^\dagger(t) \hat{\boldsymbol{\mu}} \hat{U}(t) \quad (7)$$

The induced dipole moment is defined as the change of the dipole moment after the perturbation,

$$\hat{\boldsymbol{\mu}}_{\text{ind}}(t) \equiv \hat{\boldsymbol{\mu}}(t) - \hat{\boldsymbol{\mu}}_0(t) = \hat{U}^\dagger(t) \hat{\boldsymbol{\mu}} \hat{U}(t) - e^{it\frac{\hat{H}_0}{\hbar}} \hat{\boldsymbol{\mu}} e^{-it\frac{\hat{H}_0}{\hbar}} \quad (8)$$

Substituting the first-order approximation of $\hat{U}(t)$, we have

$$\hat{\boldsymbol{\mu}}_{\text{ind}}(t) = \frac{i}{\hbar} \int_0^t dt' \mathbf{E}(t') \cdot [\hat{\boldsymbol{\mu}}_0(t), \hat{\boldsymbol{\mu}}_0(t')] \quad (9)$$

where $[\cdot, \cdot]$ is the commutator. The product of vector operators within the commutator is an **outer product (tensor product)**, the components of the resulted tensor are

$$[\hat{\mathbf{A}}, \hat{\mathbf{B}}]_{\xi\eta} = [\hat{A}_\xi, \hat{B}_\eta] = \hat{A}_\xi \hat{B}_\eta - \hat{B}_\eta \hat{A}_\xi \quad (10)$$

where $\xi, \eta = x, y, z$.

In most cases, we are dealing with the matrix element of the operators instead. For $\hat{\boldsymbol{\mu}}_{\text{ind}}(t)$ its matrix element is

$$(\boldsymbol{\mu}_{\text{ind}})_{ab} \equiv \langle a | \hat{\boldsymbol{\mu}}_{\text{ind}}(t) | b \rangle = \frac{i}{\hbar} \int_0^t dt' \mathbf{E}(t') \cdot \langle a | [\hat{\boldsymbol{\mu}}_0(t), \hat{\boldsymbol{\mu}}_0(t')] | b \rangle \quad (11)$$

Since $|a\rangle$ and $|b\rangle$ are eigenstates of the unperturbed Hamiltonian, one can show that

$$\langle a | [\hat{\boldsymbol{\mu}}_0(t), \hat{\boldsymbol{\mu}}_0(t')] | b \rangle = \langle a | [\hat{\boldsymbol{\mu}}_0(t - t'), \hat{\boldsymbol{\mu}}_0(0)] | b \rangle e^{it' \frac{E_a - E_b}{\hbar}} \quad (12)$$

In the case of $a = b$, the phase factor $e^{it' \frac{E_a - E_b}{\hbar}} = 1$. Therefore, the expectation value of induced dipole can be written as

$$\boxed{\langle \hat{\boldsymbol{\mu}}_{\text{ind}} \rangle(t) = \frac{i}{\hbar} \int_0^t dt' \mathbf{E}(t') \cdot \langle [\hat{\boldsymbol{\mu}}_0(t - t'), \hat{\boldsymbol{\mu}}_0(0)] \rangle} \quad (13)$$

The polarizability $\boldsymbol{\alpha}$ of the molecule is defined as

$$\langle \hat{\boldsymbol{\mu}}_{\text{ind}} \rangle(t) = \int_{-\infty}^{\infty} dt' \boldsymbol{\alpha}(t - t') \cdot \mathbf{E}(t') \quad (14)$$

Clearly,

$$\boldsymbol{\alpha}(t - t') = \frac{i}{\hbar} \theta(t - t') \langle [\hat{\boldsymbol{\mu}}_0(t - t'), \hat{\boldsymbol{\mu}}_0(0)] \rangle \quad (15)$$

where $\theta(t - t')$ is the Heaviside step function. Generally, it is more convenient to work in frequency space. Fourier transform of Eq. (14) yields

$$\langle \hat{\boldsymbol{\mu}}_{\text{ind}} \rangle(\omega) = \boldsymbol{\alpha}(\omega) \cdot \mathbf{E}(\omega) \quad (16)$$

Then by definition,

$$\boldsymbol{\alpha}(\omega) = \frac{i}{\hbar} \int dt \theta(t - t') \langle [\hat{\boldsymbol{\mu}}_0(t - t'), \hat{\boldsymbol{\mu}}_0(0)] \rangle e^{i\omega(t - t')} \quad (17)$$

The polarizability $\boldsymbol{\alpha}(t - t')$ and $\boldsymbol{\alpha}(\omega)$ can be viewed as the expectation value of a “polarizability operator”, $\hat{\boldsymbol{\alpha}}(t - t')$ and $\hat{\boldsymbol{\alpha}}(\omega)$,

$$\hat{\boldsymbol{\alpha}}(t - t') = \frac{i}{\hbar} \theta(t - t') [\hat{\boldsymbol{\mu}}_0(t - t'), \hat{\boldsymbol{\mu}}_0(0)] \quad (18)$$

$$\hat{\boldsymbol{\alpha}}(\omega) = \frac{i}{\hbar} \int dt \theta(t - t') [\hat{\boldsymbol{\mu}}_0(t - t'), \hat{\boldsymbol{\mu}}_0(0)] e^{i\omega(t - t')} \quad (19)$$

Let $\tau = t - t'$ and use the integral representation of the step function,

$$\theta(\tau) = \lim_{\varepsilon \rightarrow 0^+} \frac{i}{2\pi} \int_{-\infty}^{\infty} d\omega' \frac{e^{-i\omega'\tau}}{\omega' + i\varepsilon} \quad (20)$$

we have

$$\hat{\alpha}(\omega) = -\frac{1}{2\pi\hbar} \lim_{\varepsilon \rightarrow 0^+} \int d\tau \int d\omega' \frac{e^{-i(\omega' - \omega)\tau}}{\omega' + i\varepsilon} [\hat{\mu}_0(\tau), \hat{\mu}_0(0)] \quad (21)$$

Finally, we calculate the matrix element of this “polarizability operator”. We use the notation $(\alpha)_{ab}$ to denote the matrix element $\langle a | \hat{\alpha} | b \rangle$ and use $(\alpha_{\xi\eta})_{ab}$ to denote the matrix element of the components of the tensor.

$$(\alpha_{\xi\eta}(\omega))_{ab} = -\frac{1}{2\pi\hbar} \lim_{\varepsilon \rightarrow 0^+} \int d\tau \int d\omega' \frac{e^{-i(\omega' - \omega)\tau}}{\omega' + i\varepsilon} \langle a | [\hat{\mu}_0(\tau), \hat{\mu}_0(0)]_{\xi\eta} | b \rangle \quad (22)$$

Inserting the resolution of identity,

$$1 = \sum_c |c\rangle \langle c| \quad (23)$$

we have

$$\begin{aligned} \langle a | [\hat{\mu}_0(\tau), \hat{\mu}_0(0)]_{\xi\eta} | b \rangle &= \sum_c \left[\langle a | \hat{U}_0^\dagger(\tau) \hat{\mu}_\xi \hat{U}_0(\tau) | c \rangle \langle c | \hat{\mu}_\eta | b \rangle - \langle a | \hat{\mu}_\eta | c \rangle \langle c | \hat{U}_0^\dagger(\tau) \hat{\mu}_\xi \hat{U}_0(\tau) | b \rangle \right] \\ &= \sum_c \left[\langle a | \hat{\mu}_\xi | c \rangle \langle c | \hat{\mu}_\eta | b \rangle e^{-i\omega_{ac}\tau} - \langle a | \hat{\mu}_\eta | c \rangle \langle c | \hat{\mu}_\xi | b \rangle e^{i\omega_{bc}\tau} \right] \\ &= \sum_c \left[(\mu_\xi)_{ac} (\mu_\eta)_{cb} e^{-i\omega_{ac}\tau} - (\mu_\eta)_{ac} (\mu_\xi)_{cb} e^{i\omega_{bc}\tau} \right] \end{aligned} \quad (24)$$

where $(\mu)_{ac} = \langle a | \hat{\mu} | c \rangle$ is the **transition dipole moment** from state $|a\rangle$ to state $|c\rangle$,

$$\omega_{ac} = \frac{E_c - E_a}{\hbar} \quad (25)$$

is the transition frequency. Eq. (22) thus becomes

$$\begin{aligned} (\alpha_{\xi\eta}(\omega))_{ab} &= -\frac{1}{2\pi\hbar} \lim_{\varepsilon \rightarrow 0^+} \sum_c \left(\int d\tau \int d\omega' \frac{e^{-i(\omega' - \omega + \omega_{ac})\tau}}{\omega' + i\varepsilon} (\mu_\xi)_{ac} (\mu_\eta)_{cb} \right. \\ &\quad \left. - \int d\tau \int d\omega' \frac{e^{-i(\omega' - \omega - \omega_{bc})\tau}}{\omega' + i\varepsilon} (\mu_\eta)_{ac} (\mu_\xi)_{cb} \right) \end{aligned} \quad (26)$$

Integration over τ gives a Dirac delta function, thus also simplifies the integration over ω' . The final result is

$$\boxed{(\alpha_{\xi\eta}(\omega))_{ab} = -\frac{1}{\hbar} \lim_{\varepsilon \rightarrow 0^+} \sum_c \left[\frac{(\mu_\xi)_{ac} (\mu_\eta)_{cb}}{\omega - \omega_{ac} + i\varepsilon} - \frac{(\mu_\eta)_{ac} (\mu_\xi)_{cb}}{\omega + \omega_{bc} + i\varepsilon} \right]} \quad (27)$$

Setting $a = b$, we get the **Lehmann representation of the polarizability tensor** for state a . The polarizability is related to the transition dipole moments, which are intrinsically excited state properties.

2 Scattering of Light in Quantum Electrodynamics

In classical electrodynamics, the scattering of light by molecules are described as follows. The oscillating electric field of the light (with frequency ω) polarizes the electrons in the molecule, leading to an induced dipole moment. This induced dipole moment is also oscillating with frequency ω , and will then radiate electromagnetic wave with the same frequency. The radiated wave is the scattered light. This process is known as **Rayleigh scattering**. If the polarizability has some intrinsic oscillating frequency ω_0 due to the vibration and rotation of the molecule, then the scattered light will have two additional frequencies, $\omega \pm \omega_0$. This is the **Raman scattering**, which is inelastic. The radiation energy is transferred to or from the vibration or rotation DOFs of the molecule.

2.1 Transition Rate

From a quantum mechanical perspective, the scattering can be view as the transition between states

$$|a, 1_{\mathbf{k}\lambda}\rangle \rightarrow |b, 1_{\mathbf{k}'\lambda'}\rangle \quad (28)$$

where

$$|a, 1_{\mathbf{k}\lambda}\rangle = |a\rangle \otimes |1_{\mathbf{k}\lambda}\rangle \quad (29a)$$

$$|b, 1_{\mathbf{k}'\lambda'}\rangle = |b\rangle \otimes |1_{\mathbf{k}'\lambda'}\rangle \quad (29b)$$

$|a\rangle$ and $|b\rangle$ denote the states of the molecule, $|1_{\mathbf{k}\lambda}\rangle$ and $|1_{\mathbf{k}'\lambda'}\rangle$ denote the state of the photon. $|1_{\mathbf{k}\lambda}\rangle$ means one photon with wave vector $\mathbf{k}$ and polarization λ . The transition rate is given by Fermi's golden rule,

$$k_{a,1_{\mathbf{k}\lambda} \rightarrow b,1_{\mathbf{k}'\lambda'}} = \frac{2\pi}{\hbar} |V|^2 \delta(E_a - E_b + \hbar(\omega_{\mathbf{k}} - \omega_{\mathbf{k}'})) \quad (30)$$

where V is the coupling between the two states. The transition can only happen if the change of molecular energy equals the change of the photon energy.

Again we use the dipole approximation, where the interaction between the photon and the molecule is given by

$$\hat{H}_1 = -\hat{\boldsymbol{\mu}} \cdot \hat{\mathbf{E}}(\mathbf{r}) \quad (31)$$

In this equation, the electric field is treated quantum mechanically, $\hat{\mathbf{E}}(\mathbf{r})$ is the electric field operator, which is given by

$$\hat{\mathbf{E}}(\mathbf{r}) = i \sum_{\mathbf{k},\lambda} \sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2Vn_r^2\epsilon_0}} \left(\mathbf{e}_{\mathbf{k}\lambda} \hat{a}_{\mathbf{k}\lambda} e^{i\mathbf{k}\cdot\mathbf{r}} - \mathbf{e}_{\mathbf{k}\lambda}^* \hat{a}_{\mathbf{k}\lambda}^\dagger e^{-i\mathbf{k}\cdot\mathbf{r}} \right) \quad (32)$$

where $\hat{a}_{\mathbf{k}\lambda}^\dagger$ and $\hat{a}_{\mathbf{k}\lambda}$ are the creation and annihilation operators for the photon with wave vector $\mathbf{k}$ and polarization $\lambda = \pm 1$. $\mathbf{e}_{\mathbf{k}\lambda}$ is a complex unit vector that corresponds to left or right circular polarization. $\mathbf{e}_{\mathbf{k}\lambda}$ is perpendicular to $\mathbf{k}$. n_r is the refractive index of the medium. The perturbation Hamiltonian is thus

$$\hat{H}_1 = -i \sum_{\mathbf{k},\lambda} \sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2Vn_r^2\epsilon_0}} \left[(\hat{\boldsymbol{\mu}} \cdot \mathbf{e}_{\mathbf{k}\lambda}) \hat{a}_{\mathbf{k}\lambda} e^{i\mathbf{k}\cdot\mathbf{r}} - (\hat{\boldsymbol{\mu}} \cdot \mathbf{e}_{\mathbf{k}\lambda}^*) \hat{a}_{\mathbf{k}\lambda}^\dagger e^{-i\mathbf{k}\cdot\mathbf{r}} \right] \quad (33)$$

In perturbation theory, the first-order term of the coupling is given by the matrix element

$$\begin{aligned}
V^{(1)} &= \langle a, 1_{\mathbf{k}\lambda} | \hat{H}_1 | b, 1_{\mathbf{k}'\lambda'} \rangle \\
&= -i \sum_{\mathbf{k}'', \lambda''} \sqrt{\frac{\hbar\omega_{\mathbf{k}''}}{2Vn_{\text{r}}^2\varepsilon_0}} \left[((\boldsymbol{\mu})_{ab} \cdot \mathbf{e}_{\mathbf{k}''\lambda''}) \langle 1_{\mathbf{k}\lambda} | \hat{a}_{\mathbf{k}''\lambda''} | 1_{\mathbf{k}'\lambda'} \rangle e^{i\mathbf{k}'' \cdot \mathbf{r}} \right. \\
&\quad \left. - ((\boldsymbol{\mu})_{ab} \cdot \mathbf{e}_{\mathbf{k}''\lambda''}^*) \langle 1_{\mathbf{k}\lambda} | \hat{a}_{\mathbf{k}''\lambda''}^\dagger | 1_{\mathbf{k}'\lambda'} \rangle e^{-i\mathbf{k}'' \cdot \mathbf{r}} \right]
\end{aligned} \tag{34}$$

We need to calculate the matrix element of the creation and annihilation operators, $\langle 1_{\mathbf{k}\lambda} | \hat{a}_{\mathbf{k}''\lambda''} | 1_{\mathbf{k}'\lambda'} \rangle$ and $\langle 1_{\mathbf{k}\lambda} | \hat{a}_{\mathbf{k}''\lambda''}^\dagger | 1_{\mathbf{k}'\lambda'} \rangle$. Using the following properties,

$$\hat{a}_{\mathbf{k}\lambda} | 0 \rangle = 0, \quad \hat{a}_{\mathbf{k}\lambda}^\dagger | 0 \rangle = | 1_{\mathbf{k}\lambda} \rangle \tag{35a}$$

$$\hat{a}_{\mathbf{k}'\lambda'} \hat{a}_{\mathbf{k}\lambda}^\dagger = \delta_{\lambda\lambda'} \delta_{\mathbf{k}\mathbf{k}'} + \hat{a}_{\mathbf{k}\lambda}^\dagger \hat{a}_{\mathbf{k}'\lambda'} \tag{35b}$$

one can show that

$$\langle 1_{\mathbf{k}\lambda} | \hat{a}_{\mathbf{k}''\lambda''} | 1_{\mathbf{k}'\lambda'} \rangle = \langle 1_{\mathbf{k}\lambda} | \hat{a}_{\mathbf{k}''\lambda''}^\dagger | 1_{\mathbf{k}'\lambda'} \rangle = 0 \tag{36}$$

thus $V^{(1)} = 0$. **In general, if the transition changes the occupation number of more than two photon modes, the first-order coupling due to $\hat{H}_1$ is always zero.**

The second-order term of the coupling is

$$V^{(2)} = \sum_{c \neq a} \sum_P \frac{\langle a, 1_{\mathbf{k}\lambda} | \hat{H}_1 | c, P \rangle \langle c, P | \hat{H}_1 | b, 1_{\mathbf{k}'\lambda'} \rangle}{E_a + \hbar\omega_{\mathbf{k}} - E_c - E_P} \tag{37}$$

where $|P\rangle$ denote all possible photon states including the vacuum state $|0\rangle$, single mode states $|N_{\mathbf{k}\lambda}\rangle$, and multiple mode states $|N_{\mathbf{k}\lambda} M_{\mathbf{k}'\lambda'} \dots\rangle$, etc. N and M are occupation numbers of the photon modes. The energy of the photon state is denoted as E_P . **The expression of the second-order coupling indicates that the two photon processes like scattering must go through an intermediate state, $|c, P\rangle$ to occur.**

Since the matrix element of $\hat{H}_1$ is always zero if the initial and final states differ by more than two photon occupation numbers. There are only two intermediate states contribute to $V^{(2)}$, $|0\rangle$ and $|1_{\mathbf{k}\lambda} 1_{\mathbf{k}'\lambda'}\rangle$. **The intermediate state $|0\rangle$ represent a process that the incident photon is absorbed by the molecule, causing a transition from $|a\rangle$ to $|c\rangle$; then another photon is emitted by the molecule due to the transition from state $|c\rangle$ to $|b\rangle$. The intermediate state $|1_{\mathbf{k}\lambda} 1_{\mathbf{k}'\lambda'}\rangle$ represents a process that a photon is first emitted by the molecule then the incident photon is absorbed.** The expression of $V^{(2)}$ simplifies to

$$V^{(2)} = \sum_{c \neq a} \left[\frac{\langle a, 1_{\mathbf{k}\lambda} | \hat{H}_1 | c, 0 \rangle \langle c, 0 | \hat{H}_1 | b, 1_{\mathbf{k}'\lambda'} \rangle}{E_a + \hbar\omega_{\mathbf{k}} - E_c} + \frac{\langle a, 1_{\mathbf{k}\lambda} | \hat{H}_1 | c, 1_{\mathbf{k}\lambda} 1_{\mathbf{k}'\lambda'} \rangle \langle c, 1_{\mathbf{k}\lambda} 1_{\mathbf{k}'\lambda'} | \hat{H}_1 | b, 1_{\mathbf{k}'\lambda'} \rangle}{E_a + \hbar\omega_{\mathbf{k}} - E_c - \hbar\omega_{\mathbf{k}} - \hbar\omega_{\mathbf{k}'}} \right] \tag{38}$$

The matrix elements can be evaluated as

$$\langle a, 1_{\mathbf{k}\lambda} | \hat{H}_1 | c, 0 \rangle = i \sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2Vn_{\text{r}}^2\varepsilon_0}} [(\boldsymbol{\mu})_{ac} \cdot \mathbf{e}_{\mathbf{k}\lambda}^*] e^{-i\mathbf{k} \cdot \mathbf{r}} \tag{39a}$$

$$\langle c, 0 | \hat{H}_1 | b, 1_{\mathbf{k}'\lambda'} \rangle = -i \sqrt{\frac{\hbar \omega_{\mathbf{k}'}}{2V n_{\text{T}}^2 \varepsilon_0}} [(\boldsymbol{\mu})_{cb} \cdot \mathbf{e}_{\mathbf{k}'\lambda'}] e^{i\mathbf{k}' \cdot \mathbf{r}} \quad (39b)$$

$$\langle a, 1_{\mathbf{k}\lambda} | \hat{H}_1 | c, 1_{\mathbf{k}\lambda} 1_{\mathbf{k}'\lambda'} \rangle = -i \sqrt{\frac{\hbar \omega_{\mathbf{k}'}}{2V n_{\text{T}}^2 \varepsilon_0}} [(\boldsymbol{\mu})_{ac} \cdot \mathbf{e}_{\mathbf{k}'\lambda'}] e^{i\mathbf{k}' \cdot \mathbf{r}} \quad (39c)$$

$$\langle c, 1_{\mathbf{k}\lambda} 1_{\mathbf{k}'\lambda'} | \hat{H}_1 | b, 1_{\mathbf{k}'\lambda'} \rangle = i \sqrt{\frac{\hbar \omega_{\mathbf{k}}}{2V n_{\text{T}}^2 \varepsilon_0}} [(\boldsymbol{\mu})_{cb} \cdot \mathbf{e}_{\mathbf{k}\lambda}^*] e^{-i\mathbf{k} \cdot \mathbf{r}} \quad (39d)$$

Therefore

$$V^{(2)} = \frac{\hbar \sqrt{\omega_{\mathbf{k}} \omega_{\mathbf{k}'}}}{2V n_{\text{T}}^2 \varepsilon_0} e^{-i(\mathbf{k} - \mathbf{k}') \cdot \mathbf{r}} \sum_{c \neq a} \left[\frac{[(\boldsymbol{\mu})_{ac} \cdot \mathbf{e}_{\mathbf{k}\lambda}^*][(\boldsymbol{\mu})_{cb} \cdot \mathbf{e}_{\mathbf{k}'\lambda'}]}{E_a - E_c + \hbar \omega_{\mathbf{k}}} + \frac{[(\boldsymbol{\mu})_{ac} \cdot \mathbf{e}_{\mathbf{k}'\lambda'}][(\boldsymbol{\mu})_{cb} \cdot \mathbf{e}_{\mathbf{k}\lambda}^*]}{E_a - E_c - \hbar \omega_{\mathbf{k}'}} \right] \quad (40)$$

For the term in the square bracket, we can write the dot product of vectors as summation over their components,

$$\begin{aligned} \sum_{c \neq a} [\dots] &= \sum_{\xi, \eta} \sum_{c \neq a} \left(\frac{[(\mu_{\xi})_{ac} (e_{\xi})_{\mathbf{k}\lambda}^*][(\mu_{\eta})_{cb} (e_{\eta})_{\mathbf{k}'\lambda'}]}{E_a - E_c + \hbar \omega_{\mathbf{k}}} + \frac{[(\mu_{\eta})_{ac} (e_{\eta})_{\mathbf{k}'\lambda'}][(\mu_{\xi})_{cb} (e_{\xi})_{\mathbf{k}\lambda}^*]}{E_a - E_c - \hbar \omega_{\mathbf{k}'}} \right) \\ &= \sum_{\xi, \eta} (e_{\xi})_{\mathbf{k}\lambda}^* \sum_{c \neq a} \frac{1}{\hbar} \left(\frac{(\mu_{\xi})_{ac} (\mu_{\eta})_{cb}}{\omega_{\mathbf{k}} - \omega_{ac}} - \frac{(\mu_{\xi})_{cb} (\mu_{\eta})_{ac}}{\omega_{\mathbf{k}'} + \omega_{ac}} \right) (e_{\eta})_{\mathbf{k}'\lambda'} \\ &= -\mathbf{e}_{\mathbf{k}\lambda}^* \cdot (\mathbf{c})_{ab} \cdot \mathbf{e}_{\mathbf{k}'\lambda'} \end{aligned} \quad (41)$$

where we defined the **scattering tensor** $(\mathbf{c})_{ab}$. Its components are

$$(c_{\xi\eta})_{ab} = -\frac{1}{\hbar} \sum_{c \neq a} \left(\frac{(\mu_{\xi})_{ac} (\mu_{\eta})_{cb}}{\omega_{\mathbf{k}} - \omega_{ac}} - \frac{(\mu_{\xi})_{cb} (\mu_{\eta})_{ac}}{\omega_{\mathbf{k}'} + \omega_{ac}} \right) \quad (42)$$

The delta function $\delta(E_a - E_b + \hbar(\omega_{\mathbf{k}} - \omega_{\mathbf{k}'}))$ in the golden rule expression is only nonzero when

$$\omega_{\mathbf{k}'} = \omega_{\mathbf{k}} - \omega_{ab} \quad (43)$$

We can therefore replace the $\omega_{\mathbf{k}'}$ in Eq. (42) by $\omega_{\mathbf{k}} - \omega_{ab}$, which leads to

$$(c_{\xi\eta})_{ab} = -\frac{1}{\hbar} \sum_{c \neq a} \left(\frac{(\mu_{\xi})_{ac} (\mu_{\eta})_{cb}}{\omega_{\mathbf{k}} - \omega_{ac}} - \frac{(\mu_{\xi})_{cb} (\mu_{\eta})_{ac}}{\omega_{\mathbf{k}} + \omega_{bc}} \right) \quad (44)$$

On the other hand, in Eq. (27), if the frequency of the perturbation ω is not resonant with the transition frequencies, both denominators will be nonzero, and we can set $\varepsilon = 0$. As a result,

$$(\alpha_{\xi\eta}(\omega))_{ab} = -\frac{1}{\hbar} \sum_c \left(\frac{(\mu_{\xi})_{ac} (\mu_{\eta})_{cb}}{\omega - \omega_{ac}} - \frac{(\mu_{\xi})_{cb} (\mu_{\eta})_{ac}}{\omega + \omega_{bc}} \right) \quad (45)$$

$(c_{\xi\eta})_{ab}$ and $(\alpha_{\xi\eta}(\omega))_{ab}$ have almost the same expression, except that the $c = a$ term is not included in $(c_{\xi\eta})_{ab}$. **If (1) the molecular state doesn't change after the scattering (elastic scattering), i.e., $|a\rangle = |b\rangle$; or (2) the molecule has no permanent dipole on state $|a\rangle$, i.e., $(\boldsymbol{\mu})_{aa} = 0$,** then

$$\left(\frac{(\mu_{\xi})_{ac} (\mu_{\eta})_{cb}}{\omega - \omega_{ac}} - \frac{(\mu_{\xi})_{cb} (\mu_{\eta})_{ac}}{\omega + \omega_{bc}} \right) \Big|_{c=a} = 0 \quad (46)$$

the $c = a$ term can be added in the summation in $(c_{\xi\eta})_{ab}$, therefore,

$$(\mathbf{c})_{ab} = (\boldsymbol{\alpha}(\omega_{\mathbf{k}}))_{ab} \quad (47)$$

The second-order coupling thus can be written in a compact form,

$$V^{(2)} = -\frac{\hbar\sqrt{\omega_{\mathbf{k}}\omega_{\mathbf{k}'}}}{2Vn_{\text{r}}^2\varepsilon_0} (\mathbf{e}_{\mathbf{k}\lambda}^* \cdot (\boldsymbol{\alpha})_{ab} \cdot \mathbf{e}_{\mathbf{k}'\lambda'}) e^{-i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \quad (48)$$

and the transition rate becomes

$$k_{a,1_{\mathbf{k}\lambda} \rightarrow b,1_{\mathbf{k}'\lambda'}} = \frac{\pi\hbar}{2V^2n_{\text{r}}^4\varepsilon_0^2} \omega_{\mathbf{k}}\omega_{\mathbf{k}'} |\mathbf{e}_{\mathbf{k}\lambda}^* \cdot (\boldsymbol{\alpha})_{ab} \cdot \mathbf{e}_{\mathbf{k}'\lambda'}|^2 \delta(E_a - E_b + \hbar(\omega_{\mathbf{k}} - \omega_{\mathbf{k}'})) \quad (49)$$

2.2 Differential Cross Section

Eq. (49) gives the transition rate of between a single pair of states, while the total scattering rate is given by summing all possible states of the molecule and the scatter photon.

$$\begin{aligned} k_{\text{tot}} &= \sum_{a,b} f(E_a) \sum_{\mathbf{k}',\lambda'} k_{a,1_{\mathbf{k}\lambda} \rightarrow b,1_{\mathbf{k}'\lambda'}} \\ &= \sum_{a,b} f(E_a) \sum_{\mathbf{k}',\lambda'} \frac{\pi\hbar}{2V^2n_{\text{r}}^4\varepsilon_0^2} \omega_{\mathbf{k}}\omega_{\mathbf{k}'} |\mathbf{e}_{\mathbf{k}\lambda}^* \cdot (\boldsymbol{\alpha})_{ab} \cdot \mathbf{e}_{\mathbf{k}'\lambda'}|^2 \delta(E_a - E_b + \hbar(\omega_{\mathbf{k}} - \omega_{\mathbf{k}'})) \end{aligned} \quad (50)$$

where $f(E_a)$ is the Boltzmann distribution of the initial molecular states. In a finite volume $V = L_x \times L_y \times L_z$, $\mathbf{k}'$ can only take discrete values $(\frac{2\pi}{L_x}n_x, \frac{2\pi}{L_y}n_y, \frac{2\pi}{L_z}n_z)$ where $n_x, n_y, n_z = 0, \pm 1, \pm 2, \dots$. For a infinitely large volume, $V \rightarrow \infty$, the summation over $\mathbf{k}'$ can be replaced by an integral,

$$\sum_{\mathbf{k}'} \dots = \frac{V}{(2\pi)^3} \int \dots d\mathbf{k}' = \frac{Vn_{\text{r}}^3}{(2\pi c)^3} \int \dots \omega_{\mathbf{k}'}^2 d\omega_{\mathbf{k}'} d\Omega \quad (51)$$

Also note that

$$\delta(E_a - E_b + \hbar(\omega_{\mathbf{k}} - \omega_{\mathbf{k}'})) = \frac{1}{\hbar} \delta(\omega_{\mathbf{k}} - \omega_{\mathbf{k}'} - \omega_{ab}) \quad (52)$$

then

$$k_{\text{tot}} = \sum_{a,b} f(E_a) \int \frac{\sum_{\lambda'} |\mathbf{e}_{\mathbf{k}\lambda}^* \cdot (\boldsymbol{\alpha})_{ab} \cdot \mathbf{e}_{\mathbf{k}'\lambda'}|^2}{(4\pi\varepsilon_0)^2 V n_{\text{r}} c^3} \omega_{\mathbf{k}} \omega_{\mathbf{k}'}^3 \delta(\omega_{\mathbf{k}} - \omega_{\mathbf{k}'} - \omega_{ab}) d\omega_{\mathbf{k}'} d\Omega \quad (53)$$

The rate of scattering one photon within a frequency range $\omega_{\mathbf{k}'} \sim \omega_{\mathbf{k}'} + d\omega_{\mathbf{k}'}$ into the solid angle $d\Omega$ is thus

$$\frac{dk_{\text{tot}}}{d\Omega d\omega_{\mathbf{k}'}} = \sum_{a,b} f(E_a) \frac{\sum_{\lambda'} |\mathbf{e}_{\mathbf{k}\lambda}^* \cdot (\boldsymbol{\alpha})_{ab} \cdot \mathbf{e}_{\mathbf{k}'\lambda'}|^2}{(4\pi\varepsilon_0)^2 V n_{\text{r}} c^3} \omega_{\mathbf{k}} \omega_{\mathbf{k}'}^3 \delta(\omega_{\mathbf{k}} - \omega_{\mathbf{k}'} - \omega_{ab}) \quad (54)$$

The differential cross section $\frac{d\sigma}{d\Omega}$ is defined as the ratio between the rate of scattering a photon into the solid angle $d\Omega$ and the flux of the incident photon. The later is given by

$$\text{photon flux} = \frac{c}{n_{\text{r}} V} \quad (55)$$

Therefore,

$$\frac{d\sigma}{d\Omega d\omega_{\mathbf{k}'}} = \sum_{a,b} f(E_a) \frac{\sum_{\lambda'} |e_{\mathbf{k}\lambda}^* \cdot (\boldsymbol{\alpha})_{ab} \cdot \mathbf{e}_{\mathbf{k}'\lambda'}|^2}{(4\pi\epsilon_0)^2 c^4} \omega_{\mathbf{k}} \omega_{\mathbf{k}'}^3 \delta(\omega_{\mathbf{k}} - \omega_{\mathbf{k}'} - \omega_{ab}) \quad (56)$$

Eq. (56) gives the line shape of the scattering spectrum, also known as the **Raman spectrum**. The spectrum has three regions. (1) $\omega_{\mathbf{k}'} = \omega_{\mathbf{k}}$, this is the elastic Rayleigh scattering. (2) $\omega_{\mathbf{k}'} < \omega_{\mathbf{k}}$, the scattered photon shifts to lower frequency, this is called the **Stokes Raman shift**. The molecule absorbs energy of the light and transition from the ground or a lower excited state $|a\rangle$ to a higher excited state $|b\rangle$. (3) $\omega_{\mathbf{k}'} > \omega_{\mathbf{k}}$, the scattered photon shifts to higher frequency, this is called the **anti-Stokes Raman shift**. The molecule releases energy of the light and transition from a higher excited state $|a\rangle$ to the ground or a lower excited state $|b\rangle$. **Since the population of a higher excited state is much smaller than the population of the ground or lower excited states, the intensity of anti-Stokes lines are always much lower than the intensity of Stokes lines.**

If we only care about the total intensity of the scattered light instead of the distribution in frequency space, we can integrate over $\omega_{\mathbf{k}'}$ in Eq. (56), which lead to

$$\frac{d\sigma}{d\Omega} = \sum_{a,b} f(E_a) \frac{\sum_{\lambda'} |e_{\mathbf{k}\lambda}^* \cdot (\boldsymbol{\alpha})_{ab} \cdot \mathbf{e}_{\mathbf{k}'\lambda'}|^2}{(4\pi\epsilon_0)^2 c^4} \omega_{\mathbf{k}} (\omega_{\mathbf{k}} - \omega_{ab})^3 \quad (57)$$

For Rayleigh scattering, $a = b$ and $\omega_{ab} = 0$. If we further assume that a is the ground state of the molecule and $f(E_a) = 1$, then

$$\frac{d\sigma}{d\Omega} = \frac{\sum_{\lambda'} |e_{\mathbf{k}\lambda}^* \cdot \boldsymbol{\alpha} \cdot \mathbf{e}_{\mathbf{k}'\lambda'}|^2}{(4\pi\epsilon_0)^2 c^4} \omega_{\mathbf{k}}^4 \quad (58)$$

here $\boldsymbol{\alpha}$ is the ground state polarizability tensor.

In classical description, the scattered light is a spherical wave, the intensity thus depends on the distance R between the scattering center and the detector. Let I_0 be the intensity of the incident light and I be the intensity of the scattered light, then

$$\frac{I}{I_0} = \frac{1}{R^2} \frac{d\sigma}{d\Omega} \quad (59)$$

3 Coherent and Incoherent Scattering

Eq. (56) gives the differential cross section of the scattering by a single molecule. This section we will discuss the scattering by N identical, independent molecules. For the single molecule case, according to Eq. (48), the second-order approximation of the coupling is proportional to the matrix element of molecular polarizability, $(\boldsymbol{\alpha})_{ab} = \langle a | \hat{\boldsymbol{\alpha}} | b \rangle$. Strictly speaking, each molecular state should also have a phase factor $e^{i\phi}$, so is the matrix element,

$$\langle a e^{i\phi_a} | \hat{\boldsymbol{\alpha}} | b e^{i\phi_b} \rangle = (\boldsymbol{\alpha})_{ab} e^{i\Delta\phi} \quad (60)$$

where $\Delta\phi = \phi_b - \phi_a$, $(\alpha)_{ab}$ is the time-independent matrix element. With the phase factor, Eq. (48) becomes

$$V^{(2)} \propto (\mathbf{e}_{\mathbf{k}\lambda}^* \cdot (\alpha)_{ab} \cdot \mathbf{e}_{\mathbf{k}'\lambda'}) e^{i\Delta\phi} \quad (61)$$

The scattering rate and cross section only depend on $|V^{(2)}|^2$, since $|e^{i\Delta\phi}|^2 = 1$, **for single molecule case, the phase factor has no influence on the scattering rate or the cross section.**

Suppose we have N identical, independent molecules, all of them are at the same state $|a\rangle$, but have different phases. The state of the N -molecule system is

$$|a^N\rangle = \bigotimes_{j=1}^N |a^{(j)} e^{i\phi_a^{(j)}}\rangle \quad (62)$$

and the interaction between the photon and the N molecules is given by

$$\hat{H}_{1,N\text{-mol}} = - \sum_{j=1}^N \hat{\boldsymbol{\mu}}_j \cdot \hat{\mathbf{E}}(\mathbf{r}) \quad (63)$$

The coupling and its norm square for the N -molecule system are thus

$$V_{N\text{-mol}}^{(2)} \propto (\mathbf{e}_{\mathbf{k}\lambda}^* \cdot (\alpha)_{ab} \cdot \mathbf{e}_{\mathbf{k}'\lambda'}) \sum_{j=1}^N e^{i\Delta\phi^{(j)}} \quad (64a)$$

$$\left| V_{N\text{-mol}}^{(2)} \right|^2 \propto |\mathbf{e}_{\mathbf{k}\lambda}^* \cdot (\alpha)_{ab} \cdot \mathbf{e}_{\mathbf{k}'\lambda'}|^2 \left| \sum_{j=1}^N e^{i\Delta\phi^{(j)}} \right|^2 \quad (64b)$$

The phase factor $e^{i\Delta\phi^{(j)}}$ is different for every molecule.

Consider a **non-degenerated molecule**, for inelastic scattering, $a \neq b$ and $\Delta\phi^{(j)} \neq 0$.

$$\left| \sum_{j=1}^N e^{i\Delta\phi^{(j)}} \right|^2 = \sum_{j=1}^N \left| e^{i\Delta\phi^{(j)}} \right|^2 + \sum_{j \neq k}^N e^{i(\Delta\phi^{(j)} - \Delta\phi^{(k)})} = N + \sum_{j \neq k}^N e^{i\Delta\Delta\phi_{jk}} \quad (65)$$

where $\Delta\Delta\phi_{jk} \equiv \Delta\phi^{(j)} - \Delta\phi^{(k)}$. Since the phase of each molecule is independent and uniformly distributed, $\Delta\Delta\phi_{jk}$ also have a uniform distribution. The average of the second term thus equals to 0, and

$$\left| \sum_{j=1}^N e^{i\Delta\phi^{(j)}} \right|^2 = N, \quad \text{inelastic scattering } (a \neq b) \quad (66)$$

On the other hand, for elastic scattering, $a = b$ and $\Delta\phi^{(j)} = 0$, $e^{i\Delta\phi^{(j)}} = 1$. Therefore,

$$\left| \sum_{j=1}^N e^{i\Delta\phi^{(j)}} \right|^2 = N^2, \quad \text{elastic scattering } (a = b) \quad (67)$$

In summary,

$$\boxed{\left| V_{N\text{-mol}}^{(2)} \right|^2 = \begin{cases} N^2 |V^{(2)}|^2, & \text{elastic scattering } (a = b) \\ N |V^{(2)}|^2, & \text{inelastic scattering } (a \neq b) \end{cases}} \quad (68)$$

For inelastic scattering, the total differential cross section of N molecules is N -fold larger than the cross section of one molecule. Inelastic scattering is incoherent. For elastic scattering, the total differential cross section of N molecules is N^2 -fold larger than the cross section of one molecule. Elastic scattering is thus coherent. Because Rayleigh scattering is coherent, it is much stronger than the Raman scattering. On the other hand, for a degenerated system, elastic scattering can also be partially incoherent, because the molecule can undergo a transition between different states with the same energy.

4 Polarization of Scattered Light

Lastly, we discuss the polarization of the scattered light. For simplicity, we assume that the polarizability tensor is isotropic. For anisotropic molecules, we need to average over the random orientation of molecules. For an isotropic system, the polarizability tensor can be replaced by a scalar,

$$|\mathbf{e}_{\mathbf{k}\lambda}^* \cdot (\alpha)_{ab} \cdot \mathbf{e}_{\mathbf{k}'\lambda'}|^2 = \alpha_{ab}^2 |\mathbf{e}_{\mathbf{k}\lambda}^* \cdot \mathbf{e}_{\mathbf{k}'\lambda'}|^2 \equiv \alpha_{ab}^2 |\mathbf{e}_\lambda^* \cdot \mathbf{e}'_{\lambda'}|^2 \quad (69)$$

For notational simplicity, we omitted the wave vector subscript in $\mathbf{e}_{\mathbf{k}\lambda}$. Instead we use $\mathbf{e}_\lambda$ and $\mathbf{e}'_{\lambda'}$ to denote the polarization vector of the incident and scattered photon.

4.1 Circularly Polarized Incident Light

Recall that $\mathbf{e}_\lambda$ ($\lambda = \pm 1$) are complex unit vectors correspond to left and right circular polarization of the photon. If we set the coordinate system so that the wave vector of the photon is parallel to the z -axis, then

$$\mathbf{e}_1 = -\frac{1}{\sqrt{2}}(\mathbf{e}_x + i\mathbf{e}_y), \quad \mathbf{e}_{-1} = \frac{1}{\sqrt{2}}(\mathbf{e}_x - i\mathbf{e}_y) \quad (70)$$

Consider the scattering process, the wave vectors of the incident and scattered photon span the scattering plane, which is used as the xz plane in the coordinate system. The z -axis is along the wave vector of the incident photon. Let θ be the angle between $\mathbf{k}$ and $\mathbf{k}'$, then

$$\mathbf{e}_x \cdot \mathbf{e}'_x = \cos \theta, \quad \mathbf{e}_y \cdot \mathbf{e}'_y = 1, \quad \mathbf{e}_x \cdot \mathbf{e}'_y = \mathbf{e}_y \cdot \mathbf{e}'_x = 0 \quad (71)$$

Therefore,

$$\mathbf{e}_\lambda^* \cdot \mathbf{e}'_{\lambda'} = \begin{cases} \frac{1}{2}(1 + \cos \theta), & \lambda' = \lambda \\ \frac{1}{2}(1 - \cos \theta), & \lambda' \neq \lambda \end{cases} \quad (72)$$

The differential cross sections for the same and opposite circular polarization are

$$\left. \frac{d\sigma}{d\Omega} \right|_{\lambda'=\lambda} = \sum_{a,b} f(E_a) \frac{\alpha_{ab}^2}{(4\pi\epsilon_0)^2 c^4} \omega_{\mathbf{k}} (\omega_{\mathbf{k}} - \omega_{ab})^3 \frac{(1 + \cos \theta)^2}{4} \quad (73a)$$

$$\left. \frac{d\sigma}{d\Omega} \right|_{\lambda' \neq \lambda} = \sum_{a,b} f(E_a) \frac{\alpha_{ab}^2}{(4\pi\epsilon_0)^2 c^4} \omega_{\mathbf{k}} (\omega_{\mathbf{k}} - \omega_{ab})^3 \frac{(1 - \cos \theta)^2}{4} \quad (73b)$$

and the total differential cross section is

$$\frac{d\sigma}{d\Omega} = \sum_{a,b} f(E_a) \frac{\alpha_{ab}^2}{(4\pi\epsilon_0)^2 c^4} \omega_{\mathbf{k}} (\omega_{\mathbf{k}} - \omega_{ab})^3 \frac{1 + \cos^2 \theta}{2} \quad (74)$$

The ratio between the intensity of scattered light with opposite and same polarization is

$$\boxed{\frac{I_{\text{op}}}{I_{\text{sp}}} = \left(\frac{1 - \cos \theta}{1 + \cos \theta} \right)^2} \quad (75)$$

For a circularly polarized incident light, when $\theta = 0$, $I_{\text{op}} = 0$, the forward scattered light retains the polarization of the incident light. When $\theta = \pi$, $I_{\text{sp}} = 0$, the back scattered light has completely flipped polarization. Along other directions, the scattered light can have both same or opposite polarization as the incident light. Their superposition leads to elliptical polarization or linear polarization.

4.2 Unpolarized Incident Light: Depolarization Ratio

In all derivations above, we have chosen the left and right circular polarization as the basis for the polarization states of the incident and scattered photon. Other polarization states like the elliptical polarization or linear polarization are the superposition of these circular polarization states:

$$\mathbf{e} = c_1 \mathbf{e}_1 + c_{-1} \mathbf{e}_{-1} \quad (76)$$

$$|1_{\mathbf{k}, \mathbf{e}}\rangle = c_1 |1_{\mathbf{k}, 1}\rangle + c_{-1} |1_{\mathbf{k}, -1}\rangle \quad (77)$$

where $\mathbf{e}$ is the polarization vector (of any kind) of the light. $c_1^2 + c_{-1}^2 = 1$. Using these relation, we can calculate the transition rate between photons with other kinds of polarization,

$$|a, 1_{\mathbf{k}, \mathbf{e}}\rangle \rightarrow |b, 1_{\mathbf{k}', \mathbf{e}'}\rangle \quad (78)$$

and the corresponding cross section. The result is just

$$\boxed{\frac{d\sigma}{d\Omega} = \sum_{a,b} f(E_a) \frac{|\mathbf{e}^* \cdot (\boldsymbol{\alpha})_{ab} \cdot \mathbf{e}'|^2}{(4\pi\epsilon_0)^2 c^4} \omega_{\mathbf{k}} (\omega_{\mathbf{k}} - \omega_{ab})^3} \quad (79)$$

Compared to Eq. (57), the only difference is that $\mathbf{e}_{\mathbf{k}\lambda}$ and $\mathbf{e}_{\mathbf{k}'\lambda'}$ being replaced by $\mathbf{e}$ and $\mathbf{e}'$. The summation over λ' is discarded because we only care about the scattered light with a given polarization $\mathbf{e}'$. If the molecular polarizability is isotropic, then

$$\frac{d\sigma}{d\Omega} = \sum_{a,b} f(E_a) \frac{\alpha_{ab}^2 |\mathbf{e}^* \cdot \mathbf{e}'|^2}{(4\pi\epsilon_0)^2 c^4} \omega_{\mathbf{k}} (\omega_{\mathbf{k}} - \omega_{ab})^3 \quad (80)$$

Consider linearly polarized scattered light, any $\mathbf{e}'$ can be decomposed into two orthogonal vectors $\mathbf{e}'_{\parallel}$ and $\mathbf{e}'_{\perp}$. **$\mathbf{e}'_{\parallel}$ refers to that the polarization vector of the scatter light is parallel to the scattering plane (a plane spanned by $\mathbf{k}$ and $\mathbf{k}'$), while $\mathbf{e}'_{\perp}$ refers to that the polarization**

vector of the scatter light is perpendicular to the scattering plane. We set the scattering plane as the xz plane. By definition, $\mathbf{e}'_{\parallel}$ is parallel to the xz plane and $\mathbf{e}'_{\perp}$ is parallel to the y axis. The angle between $\mathbf{k}$ and $\mathbf{k}'$ is denoted as θ . $\phi \in [0, 2\pi]$ is the angle between $\mathbf{e}$ and the x axis. Then

$$|\mathbf{e} \cdot \mathbf{e}'_{\parallel}|^2 = \cos^2 \theta \cos^2 \phi \quad (81a)$$

$$|\mathbf{e} \cdot \mathbf{e}'_{\perp}|^2 = \sin^2 \phi \quad (81b)$$

If the incident light is unpolarized, ϕ is uniformly distributed within the interval $[0, 2\pi]$. Taking the average with respect to ϕ , we have

$$\langle |\mathbf{e} \cdot \mathbf{e}'_{\parallel}|^2 \rangle_{\phi} = \frac{1}{2} \cos^2 \theta \quad (82a)$$

$$\langle |\mathbf{e} \cdot \mathbf{e}'_{\perp}|^2 \rangle_{\phi} = \frac{1}{2} \quad (82b)$$

The differential cross sections for $\mathbf{e}'_{\parallel}$ and $\mathbf{e}'_{\perp}$ components are thus

$$\left. \frac{d\sigma}{d\Omega} \right|_{\parallel} = \sum_{a,b} f(E_a) \frac{\alpha_{ab}^2}{2(4\pi\epsilon_0)^2 c^4} \omega_{\mathbf{k}} (\omega_{\mathbf{k}} - \omega_{ab})^3 \cos^2 \theta \quad (83a)$$

$$\left. \frac{d\sigma}{d\Omega} \right|_{\perp} = \sum_{a,b} f(E_a) \frac{\alpha_{ab}^2}{2(4\pi\epsilon_0)^2 c^4} \omega_{\mathbf{k}} (\omega_{\mathbf{k}} - \omega_{ab})^3 \quad (83b)$$

and the total differential cross section is

$$\boxed{\frac{d\sigma}{d\Omega} = \sum_{a,b} f(E_a) \frac{\alpha_{ab}^2}{(4\pi\epsilon_0)^2 c^4} \omega_{\mathbf{k}} (\omega_{\mathbf{k}} - \omega_{ab})^3 \frac{1 + \cos^2 \theta}{2}} \quad (84)$$

For Rayleigh scattering specifically, differential cross sections for $\mathbf{e}'_{\parallel}$ and $\mathbf{e}'_{\perp}$ components are

$$\left. \frac{d\sigma}{d\Omega} \right|_{\parallel} = \frac{\alpha^2 \omega_{\mathbf{k}}^4}{2(4\pi\epsilon_0)^2 c^4} \cos^2 \theta \quad (85a)$$

$$\left. \frac{d\sigma}{d\Omega} \right|_{\perp} = \frac{\alpha^2 \omega_{\mathbf{k}}^4}{2(4\pi\epsilon_0)^2 c^4} \quad (85b)$$

and the total differential cross section is

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{\alpha^2 \omega_{\mathbf{k}}^4}{(4\pi\epsilon_0)^2 c^4} \frac{1 + \cos^2 \theta}{2}} \quad (86)$$

The **depolarization ratio** is defined as the ratio between the intensity of scattered light with polarization $\mathbf{e}'_{\parallel}$ and $\mathbf{e}'_{\perp}$

$$\boxed{\frac{I_{\parallel}}{I_{\perp}} = \cos^2 \theta} \quad (87)$$

If the incident light is unpolarized, then the scattered light will be polarized. If the scattering medium is isotropic, at $\theta = \pi/2$, the scattered light will be linearly polarized, and the polarization direction is perpendicular to the scattering plane.